

1. (4 points per part) For parts (a)–(c), consider the plane $\mathcal{P} : 2x + y + 3z = 2$ and the line $L : x = 1 - 4t, y = 2t, z = 5 + t$.

- (a) Find the acute angle between the line L and the line $y = 2, z = 6$.

The line L has a direction vector $\mathbf{v} = \langle -4, 2, 1 \rangle$.

The line $y = 2, z = 6$ is parallel to the x -axis, it has a direction vector $\mathbf{i} = \langle 1, 0, 0 \rangle$.

$$\cos^{-1} \left(\frac{\mathbf{v} \cdot \mathbf{i}}{|\mathbf{v}||\mathbf{i}|} \right) = \frac{-4}{\sqrt{21}}$$

The acute angle is $\theta = \pi - \cos^{-1} \left(\frac{-4}{\sqrt{21}} \right)$ or $\cos^{-1} \left(\frac{4}{\sqrt{21}} \right)$.

- (b) Find the equation of the plane that contains the line L and is perpendicular to the plane $\mathcal{P}$.

Since the unknown plane contains the line L , it contains any point on L , for example, at $t = 0$, $P_0(1, 0, 5)$.

The normal vector of the unknown plane is perpendicular to both the line vector $\mathbf{v} = \langle -4, 2, 1 \rangle$ and the normal vector of the given plane $\mathbf{n} = \langle 2, 1, 3 \rangle$.

Therefore a normal vector of the unknown plane is

$$\mathbf{v} \times \mathbf{n} = \langle -4, 2, 1 \rangle \times \langle 2, 1, 3 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = \langle 5, 14, -8 \rangle$$

The plane equation is $5(x - 1) + 14(y - 0) - 8(z - 5) = 0$. (answer not unique)

- (c) Find two points A and B on the plane $\mathcal{P}$ such that the vector $\overrightarrow{AB}$ is perpendicular to the line L .

A vector $\overrightarrow{AB}$ lying on (parallel to) the plane $\mathcal{P}$ is perpendicular to the normal vector of the plane, since $\overrightarrow{AB}$ is also perpendicular to the line L , we may find

$$\overrightarrow{AB} = \mathbf{v} \times \mathbf{n} = \langle -4, 2, 1 \rangle \times \langle 2, 1, 3 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = \langle 5, 14, -8 \rangle \quad (\text{already found in (b)})$$

Then it suffices to find any point A on the plane $\mathcal{P}$ and then find the point B such that $\overrightarrow{AB} = \langle 5, 14, -8 \rangle$. For example,

$A(1, 0, 0)$ and $B(6, 14, -8)$ (answer not unique)

2. (4 points per part) For parts (a)–(c), a particle is moving along a space curve with initial position vector $\mathbf{r}(0) = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$. The velocity vector of the particle is

$$\mathbf{v}(t) = 3 \cos(t) \mathbf{i} + 4t \mathbf{j} + \sin(t) \mathbf{k}$$

- (a) Find the position vector function $\mathbf{r}(t)$.

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt = \langle 3 \sin(t) + C_1, 2t^2 + C_2, -\cos(t) + C_3 \rangle.$$

$$\text{Since } \mathbf{r}(0) = \langle C_1, C_2, -1 + C_3 \rangle = 2\mathbf{i} + \mathbf{j} - \mathbf{k}, \quad C_1 = 2, C_2 = 1, C_3 = 0,$$

$$\mathbf{r}(t) = \langle 3 \sin(t) + 2, 2t^2 + 1, -\cos(t) \rangle.$$

- (b) Find the equation of a surface that contains the space curve $\mathbf{r}(t)$.

$$\text{Since } \sin(t) = \frac{x-2}{3}, \cos(t) = -z,$$

$$\mathbf{r}(t) \text{ is contained in the cylinder } \left(\frac{x-2}{3} \right)^2 + z^2 = 1.$$

- (c) Find the curvature of the space curve $\mathbf{r}(t)$ at $t = \pi$.

$$\mathbf{r}'(\pi) = \mathbf{v}(\pi) = \langle -3, 4\pi, 0 \rangle,$$

$$\mathbf{r}''(t) = \mathbf{v}'(t) = \langle -3 \sin(t), 4, \cos(t) \rangle, \quad \mathbf{r}''(\pi) = \langle 0, 4, -1 \rangle,$$

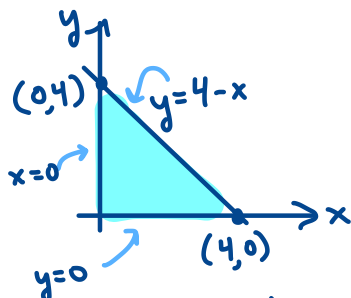
$$\mathbf{r}'(\pi) \times \mathbf{r}''(\pi) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 4\pi & 0 \\ 0 & 4 & -1 \end{vmatrix} = \langle -4\pi, -3, -12 \rangle,$$

$$\kappa(\pi) = \frac{|\mathbf{r}'(\pi) \times \mathbf{r}''(\pi)|}{|\mathbf{r}'(\pi)|^3} = \frac{|\langle -4\pi, -3, -12 \rangle|}{|\langle -3, 4\pi, 0 \rangle|^3} = \frac{\sqrt{153 + 16\pi^2}}{(9 + 16\pi^2)^{3/2}}.$$

3. (12 points) Let f be the function given by:

$$f(x, y) = x^2 - y^2 - 2xy + 4y,$$

and let A , B , and C be the points $(0, 0)$, $(0, 4)$ and $(4, 0)$. Find the global minimum and maximum of f in the filled triangle ABC .



Critical points: $f_x(x, y) = 2x - 2y = 0 \rightarrow x = y$
 $f_y(x, y) = -2y - 2x + 4 = 0 \rightarrow 4x = 4 \rightarrow x = y = 1$
 $(1, 1)$ is a crit. point.

Left edge: $x = 0$
 $f(0, y) = -y^2 + 4y$
 $f'(y) = -2y + 4 = 0 \rightarrow y = 2$ Check $(0, 2)$

Bottom edge: $y = 0$
 $f(x, 0) = x^2$
 $f'(x) = 2x = 0 \rightarrow x = 0$ Check $(0, 0)$

Top-right edge: $y = 4 - x$
 $f(x, 4 - x) = x^2 - (4 - x)^2 - 2x(4 - x) + 4(4 - x) = 2x^2 - 4x$
 $f'(x) = 4x - 4 = 0 \rightarrow x = 1$, Check $(1, 3)$.

Also check endpoints $(0, 0)$, $(0, 4)$, $(4, 0)$.

Plug in:

$$f(0, 0) = 0$$

$$f(1, 1) = 2$$

$$f(0, 4) = 0$$

$$f(4, 0) = 16 \leftarrow \text{max}$$

$$f(0, 2) = 4$$

$$f(1, 3) = -2 \leftarrow \text{min}$$

4. (12 points) Find a point on the surface of equation $z = x^2 - y^2$ whose tangent plane is perpendicular to the line of equation

$$x = t$$

$$y = 3t$$

$$z = -2t.$$

direction $\langle 1, 3, -2 \rangle$

$$\frac{\partial z}{\partial x} = 2x$$

$$\frac{\partial z}{\partial y} = -2y$$

Normal vector $\langle 2x, -2y, -1 \rangle$

So $\langle 1, 3, -2 \rangle$ is parallel to $\langle 2x, -2y, -1 \rangle$

$$\langle 1, 3, -2 \rangle = c \langle 2x, -2y, -1 \rangle$$

$c = 2$

$$\langle 1, 3, -2 \rangle = \langle 4x, -4y, -2 \rangle$$

$$x = \frac{1}{4}, \quad y = \frac{-3}{4}, \quad z = x^2 - y^2 = \frac{-1}{2}$$

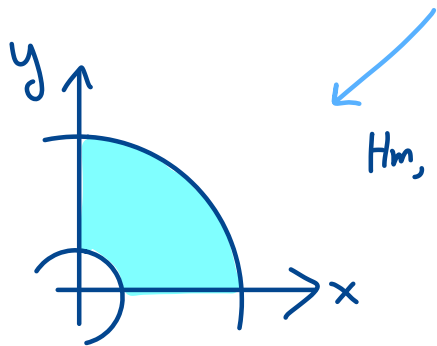
So the point of tangency is $\left(\frac{1}{4}, \frac{-3}{4}, \frac{-1}{2} \right)$

5. (12 points) Find the volume of the solid below the surface

$$z = \frac{2x + y + 1}{\sqrt{x^2 + y^2}}$$

and above the region in the xy -plane

$$D = \{(x, y) \mid x \geq 0, y \geq 0, 1 \leq x^2 + y^2 \leq 4\}.$$



Hm, looks polar.

$$\int_0^{\frac{\pi}{2}} \int_1^2 \frac{2r \cos \theta + r \sin \theta + 1}{r} dr d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_1^2 (2r \cos \theta + r \sin \theta + 1) dr d\theta = \int_0^{\frac{\pi}{2}} \left(r^2 \cos \theta + \frac{1}{2} r^2 \sin \theta + r \right) \Big|_{r=1}^{r=2} d\theta$$

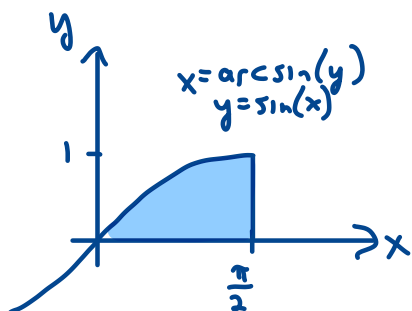
$$= \int_0^{\frac{\pi}{2}} \left(3 \cos \theta + \frac{3}{2} \sin \theta + 1 \right) d\theta = \left(3 \sin \theta - \frac{3}{2} \cos \theta + \theta \right) \Big|_0^{\frac{\pi}{2}}$$

$$= \left(3 + \frac{\pi}{2} \right) - \left(-\frac{3}{2} \right)$$

$$= \boxed{\frac{9 + \pi}{2}}$$

6. (12 points) Evaluate the integral

$$\int_0^1 \int_{\arcsin y}^{\pi/2} \cos x \sqrt{1 + \cos^2 x} \, dx \, dy.$$



Impossible.

Must reverse order of integration.

$$\int_0^{\pi/2} \int_0^{\sin(x)} \cos(x) \sqrt{1 + \cos^2 x} \, dy \, dx$$

$$= \int_0^{\pi/2} \left(y \cos x \sqrt{1 + \cos^2 x} \right) \Big|_{y=0}^{y=\sin x} \, dx$$

$$= \int_0^{\pi/2} \sin x \cos x \sqrt{1 + \cos^2 x} \, dx$$

$$u = 1 + \cos^2 x$$

$$du = -2 \sin x \cos x \, dx$$

$$= \int_2^1 \frac{-1}{2} \sqrt{u} \, du = \frac{-1}{3} u^{3/2} \Big|_2^1 = \frac{-1}{3} (1 - 2\sqrt{2})$$

$$= \frac{2\sqrt{2} - 1}{3}$$

7. For parts (a)–(c), let $f(x) = \int_2^x \cos(\pi t^2) dt$.

(a) (5 points) Find the second Taylor polynomial, $T_2(x)$, for $f(x)$ based at $b = 2$.

$$\begin{aligned} f(x) &= \int_2^x \cos(\pi t^2) dt & f(2) &= 0 \\ f'(x) &= \cos(\pi x^2) & f'(2) &= 1 \\ f''(x) &= -2\pi x \sin(\pi x^2) & f''(2) &= 0 \end{aligned}$$

$$T_2(x) = (x-2)$$

(b) (3 points) Use your answer to part (a) to approximate $\int_2^{2.1} \cos(\pi t^2) dt$.

$$\int_2^{2.1} \cos(\pi t^2) dt = f(2.1) \approx T_2(2.1) = 2.1 - 2 = 0.1$$

(c) (6 points) Find an upper bound on the error for your answer from part (b).

(Note: For full credit, you do not need to find a tight upper bound, but you must justify your answer with Taylor's inequality.)

$$\begin{aligned} f'''(x) &= -2\pi \sin(\pi x^2) - 4\pi^2 x^2 \cos(\pi x^2) \\ |f'''(x)| &\leq \underbrace{|2\pi \sin(\pi x^2)|}_{\leq 2\pi} + \underbrace{|4\pi^2 x^2 \cos(\pi x^2)|}_{\leq 17.64\pi^2 \text{ on } [2, 2.1]} \leq 2\pi + 17.64\pi^2 \end{aligned}$$

(other answers are possible, but note that neither $|f'''(2)|$ nor $|f'''(2.1)|$ is an upper bound)

$$\text{So } |T_2(x) - f(x)| \leq \frac{1}{6} (2\pi + 17.64\pi^2) (0.1)^3 \approx 0.03006$$

8. For this problem, you may use the following basic Taylor series:

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k, \quad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \quad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}, \quad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

- (a) (6 points) Find the Taylor series for $f(x) = \frac{x^3}{1+x^4} - 3x \sin(x^2)$ based at $b = 0$. Express your answer using $\sum$ -notation.

$$\begin{aligned} \frac{1}{1-x} &= \sum_{k=0}^{\infty} x^k \\ x \rightarrow -x^4 \downarrow \\ \frac{1}{1+x^4} &= \sum_{k=0}^{\infty} (-1)^k x^{4k} \\ \frac{x^3}{1+x^4} &= \sum_{k=0}^{\infty} (-1)^k x^{4k+3} \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} \\ x \rightarrow x^2 \downarrow \\ \sin(x^2) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{4k+2}}{(2k+1)!} \\ -3x \sin(x^2) &= \sum_{k=0}^{\infty} \frac{-3(-1)^k x^{4k+3}}{(2k+1)!} \\ \frac{x^3}{1+x^4} - 3x \sin(x^2) &= \sum_{k=0}^{\infty} (-1)^k \left(1 - \frac{3}{(2k+1)!} \right) x^{4k+3} \end{aligned}$$

- (b) (3 points) Find the open interval of convergence for the series you found in (a).

$$\begin{aligned} \frac{1}{1-x} \text{ converges for } -1 < x < 1 \\ x \rightarrow -x^4 \downarrow \\ -1 < -x^4 < 1 \\ \text{still } (-1, 1) \\ \sin(x) \text{ converges for all } x \end{aligned}$$

- (c) (5 points) Find $f^{(2023)}(0)$, i.e. the 2023rd derivative of f at 0.

$$x^{2023} \text{ term of } \text{occurs at } 4k+3=2023 \rightarrow k=505$$

$$\frac{f^{(2023)}(0)}{2023!} x^{2023} = (-1)^{505} \left(1 - \frac{3}{1011!} \right) x^{2023}$$

$$f^{(2023)}(0) = - \left(1 - \frac{3}{1011!} \right) (2023!)$$